

TCP/IP Model

1. What is the TCP/IP Protocol Suite?

The **Internet Protocol Suite** — commonly called **TCP/IP** — is the family of protocols that makes communication over the Internet and other networks possible. Well-known protocols in this suite include IP, TCP, UDP, HTTP, and DNS.

Key Terms:

- **Protocol:** A set of rules defining how data should be communicated between devices.
- **Standard:** An agreed-upon specification describing how a protocol or technology should work.
- **Vendor-neutral:** Standards not owned by any single company — any device can implement them.

2. Brief History of TCP/IP

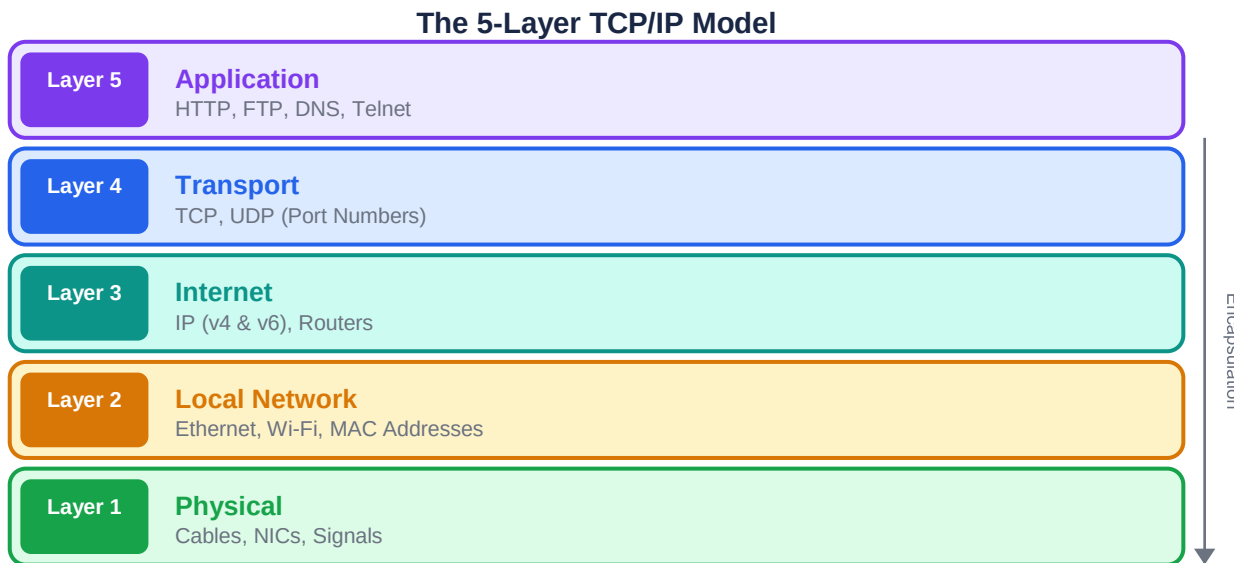
Year	Event
1960s	US DoD ARPA funds ARPANET research
1969	ARPANET goes online, connecting university mainframes
1974	Vint Cerf & Bob Kahn begin developing TCP
1983	ARPANET fully switches to TCP/IP (Jan 1st)
Today	TCP/IP dominates all modern networking

3. Standards Organizations

Organization	Full Name	What They Define
IEEE	Institute of Electrical and Electronics Engineers	Ethernet (802.3), Wi-Fi (802.11) — physical & link specs
IETF	Internet Engineering Task Force	TCP, IP, UDP, HTTP, DNS — publishes open RFCs

■ **Tip:** RFCs (Requests for Comments) are freely available online — RFC 791 defines IP and is still the current standard.

4. The 5-Layer TCP/IP Model



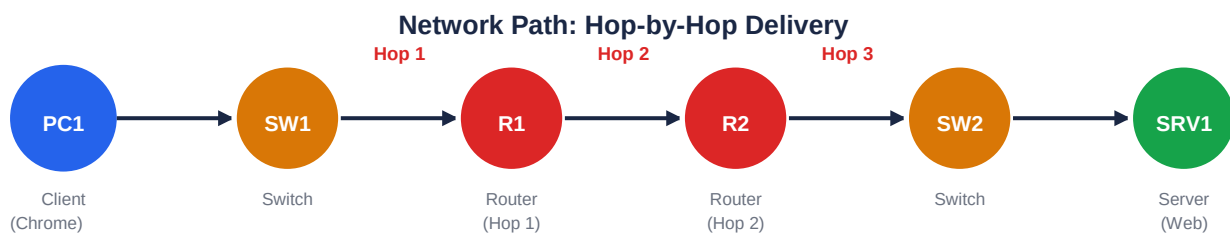
5. Layer-by-Layer Details

Layer Roles & Key Protocols

Layer	Role	Key Protocols
Application	App-to-app data exchange	HTTP, FTP, DNS, Telnet
Transport	End-to-end via port numbers	TCP, UDP
Internet	Host-to-host via IP addresses	IPv4, IPv6
Local Network	Hop-to-hop via MAC addresses	Ethernet, Wi-Fi
Physical	Bit transmission as signals	UTP, Fiber, Wi-Fi radio

Layer 1 — Physical	Sends/receives bits as signals over physical media. Covers cables, connectors, signal levels, and link speeds. Examples: UTP copper cables, fiber-optic cables, Wi-Fi radios, NICs .
Layer 2 — Local Network	Provides hop-to-hop delivery within a local network (LAN). Uses MAC addresses to identify interfaces. A hop = one step between routers or hosts. Switches extend a LAN but do NOT count as hops. Key protocols: Ethernet, Wi-Fi .
Layer 3 — Internet	Provides end-to-end delivery between hosts across multiple networks (internetworks). Uses IP addresses (IPv4 & IPv6) and routers . Think: 'IP addresses and routers'. A host = any device that can send/receive data.
Layer 4 — Transport	Provides end-to-end communication between application processes . Uses port numbers to identify which process to send data to (e.g., port 80 = web server, port 21 = file server). Key protocols: TCP (reliable) and UDP (fast/unreliable).
Layer 5 — Application	Responsible for creating and interpreting data for specific applications. Includes protocols for communication between application processes. Examples: HTTP, FTP, Telnet, DNS .

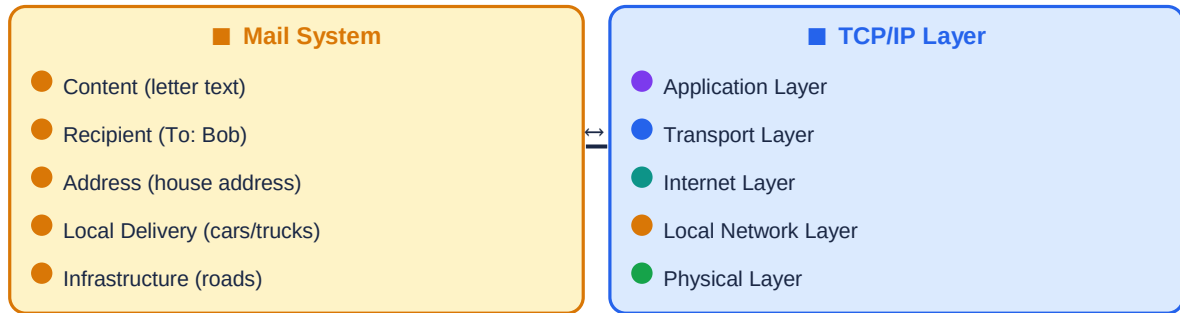
6. How Data Travels: Hops



Switches extend the LAN and are **not counted as hops**. Only routers (and source/destination hosts) define hop boundaries.

7. Mail System Analogy

Mail System Analogy ↔ TCP/IP Model



8. Quick-Reference Cheat Sheet

Concept	Remember As
Protocol	Language computers use to communicate
Standard	Agreed spec so all vendors can interoperate
IEEE	Defines Ethernet (802.3) & Wi-Fi (802.11)
IETF	Defines TCP, IP, UDP, HTTP, DNS via RFCs
Physical (L1)	Bits → signals (copper, fiber, radio)
Local Network (L2)	MAC addresses + switches → hop-to-hop
Internet (L3)	IP addresses + routers → end-to-end hosts
Transport (L4)	Port numbers → end-to-end processes
Application (L5)	App protocols (HTTP, FTP, DNS)
TCP/IP adopted	January 1, 1983 on ARPANET
Vint Cerf & Bob Kahn	Fathers of TCP/IP (developed 1974)